

Neural World Model & Predictive Robot Intelligence

Technical Report — learned dynamics, multi-future imagination, and risk-aware model-predictive control for mobile robot navigation among moving obstacles

Abstract

We ask whether a mobile robot can learn a model of its world from its own experience and use that model to predict future states before making decisions. We train a neural world model comprising (i) a learned robot dynamics model and (ii) a rotation-invariant, history-conditioned obstacle motion predictor, and embed both inside a sampling-based model-predictive planner (MPPI) that imagines 220 candidate futures at 10 Hz. Against identical-cost baselines, the learned predictor reduces 3-second obstacle forecasting error by 36% in distribution and 58% under distribution shift (ADE), and the imagination planner achieves 100% success with zero collisions where a reactive controller collides in 33% of episodes. We then upgrade the model to a 5-member bootstrapped ensemble: its disagreement is a calibrated epistemic-uncertainty signal ($r = 0.49$ against realized error), and planning over multiple imagined futures reduces collisions by 40% in a stress environment relative to a single imagined future. All experiments are seeded, CPU-reproducible in minutes, and containerized.

1. Method

Environment. A 20×20 m arena with a unicycle robot (state x, y, θ ; action v, ω) and K moving circular obstacles following noisy constant-speed, constant-turn-rate dynamics with wall bounces — motion that is smooth but non-linear, so straight-line extrapolation degrades with horizon.

World model. The robot dynamics model is an MLP mapping commanded (v, ω) to body-frame displacement, trained on 5,500 logged transitions ($\text{MSE} \approx 2 \times 10^{-4}$). The obstacle model maps the last 6 velocity vectors — rotated into the obstacle's current heading frame — to the next velocity, and is rolled out autoregressively. Rotation invariance is the key inductive bias: the model learns the turn-rate/speed relationship once, independent of world-frame direction, which drives the observed distribution-shift robustness. Wall-bounce discontinuities are filtered from training. An ensemble of 5 bootstrapped members provides epistemic uncertainty via inter-member disagreement (PETS-style).

Planner. MPPI samples 220 action sequences (1.8 s horizon), rolls each out inside the learned model against imagined obstacle futures, and scores collision, proximity, goal progress, effort, and smoothness. Critically, the MPPI baseline and Neural-MPC share the identical optimizer and cost; only the imagination model differs, isolating the world model's contribution. The risk-aware variant scores each action by CVaR over the 5 ensemble futures.

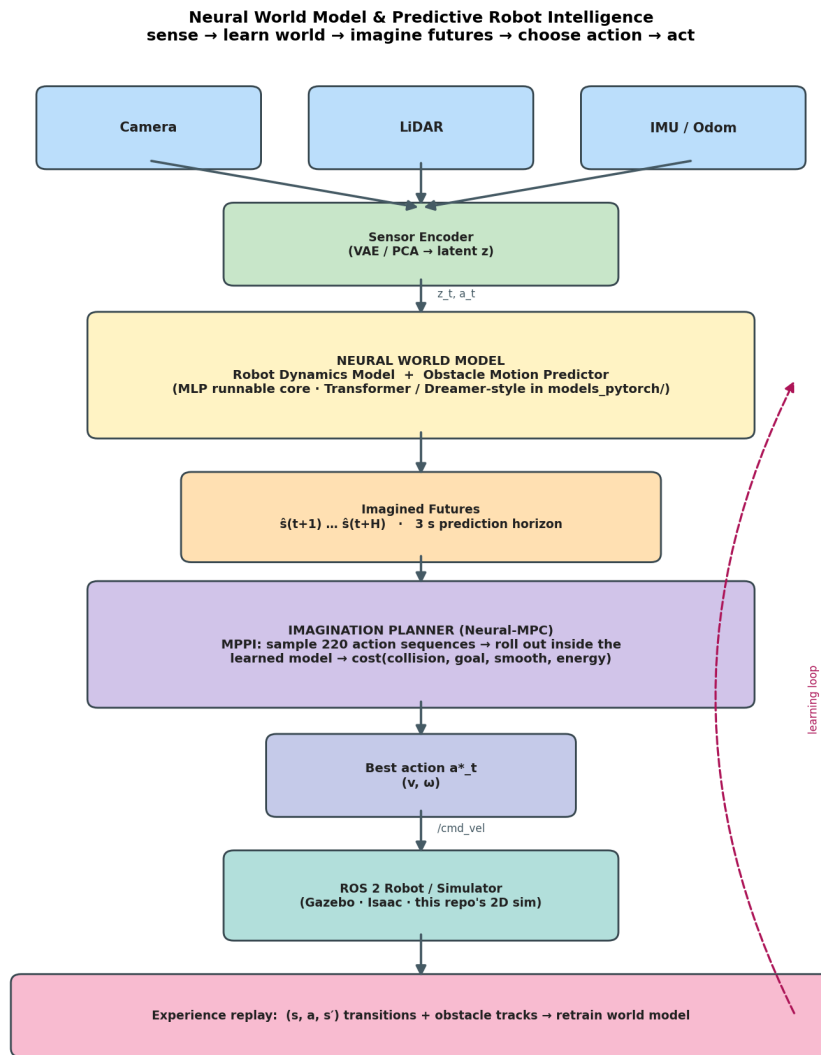


Fig. 1 — System architecture and learning loop.

2. Experiments and Results

2.1 Test 1 — Dynamic obstacle prediction (ADE/FDE, 3 s, 40 episodes)

Predictor	ADE in-dist [m]	FDE in-dist [m]	ADE unseen [m]	FDE unseen [m]
Constant velocity	0.190	0.522	0.448	1.231
Neural world model	0.121	0.335	0.187	0.522
Improvement	-36%	-36%	-58%	-58%

Test 1 — Imagined obstacle futures (3 s horizon)

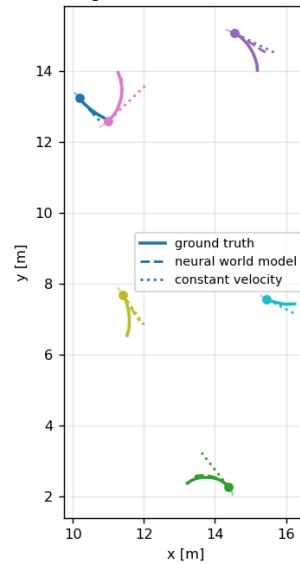


Fig. 2 — Imagined futures (dashed) track curved ground truth; constant velocity (dotted) diverges.

2.2 Test 2/3 — Planning benchmark and generalization (30 episodes each)

Planner	Success in-dist	Collision in-dist	Success unseen	Collision unseen
Reactive	66.7%	33.3%	70.0%	30.0%
DWA	100.0%	0.0%	96.7%	3.3%
MPPI	93.3%	0.0%	93.3%	0.0%
Neural-MPC	100.0%	0.0%	93.3%	0.0%

Test 2 — Planner benchmark (training environment)

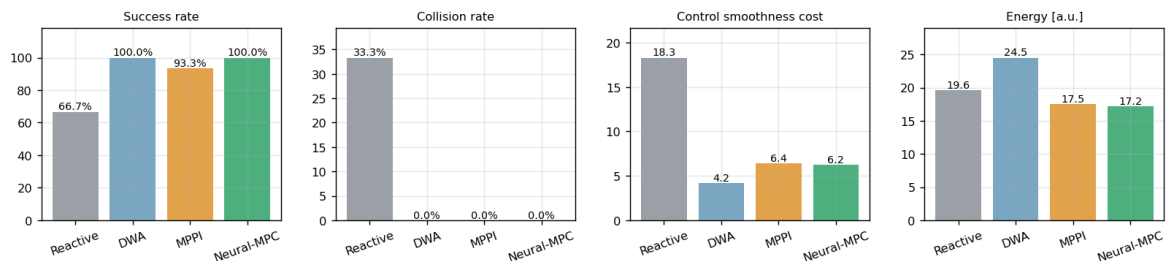


Fig. 3 — Planner benchmark, training environment.

2.3 Test 4 — Uncertainty calibration

Across 720 obstacle tracks in the stress environment, ensemble disagreement correlates with realized prediction error at $r = 0.49$: the model's confidence signal is informative rather than noise.

Test 4 — Epistemic uncertainty from the ensemble world model

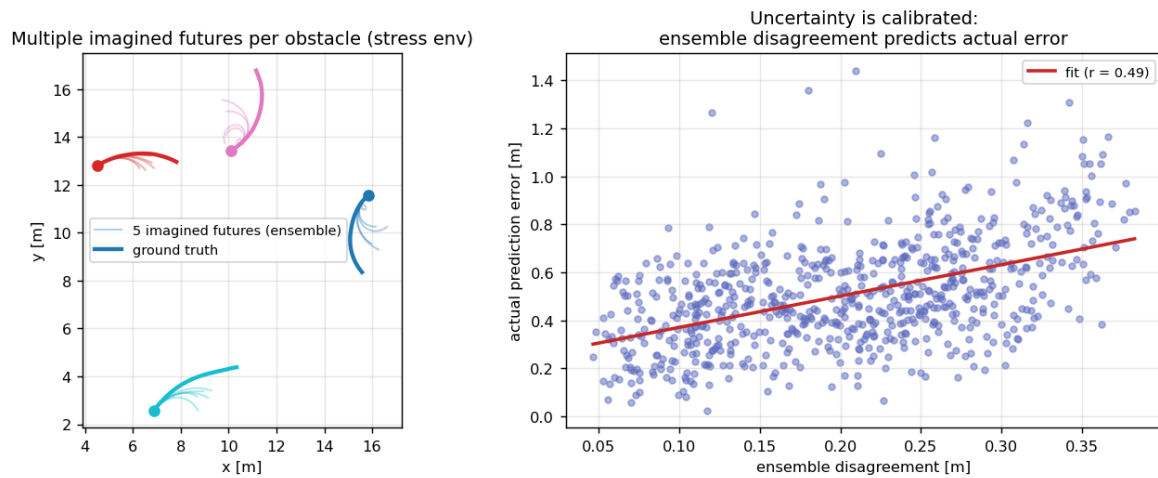


Fig. 4 — Left: five imagined futures per obstacle. Right: disagreement vs error.

2.4 Test 5 — Multi-future imagination under stress (12 fast noisy obstacles, 20 episodes)

Planner	Imagined futures	Success	Collision
Single world model	1	70%	25%
Ensemble mean	5 (averaged)	75%	15%
Ensemble CVaR	5 (worst-case)	70%	15%

Test 5 — Stress environment (12 fast, noisy obstacles): deterministic vs risk-aware imagination

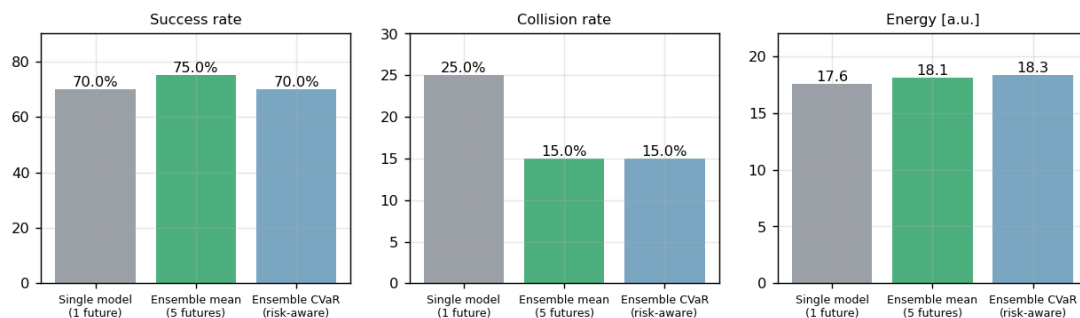


Fig. 5 — One vs five imagined futures: collisions drop 25% to 15%.

Negative results (reported by design). Full CVaR pessimism matched the ensemble mean on safety but paid a small success penalty through over-caution; an uncertainty-inflated safety margin variant performed worse, as inflated margins in a dense field leave no admissible paths. With 5 members, worst-case statistics are noisy; larger ensembles or a mean + $\beta \cdot \text{std}$ risk term are the indicated next step.

3. Reproducibility

All experiments are deterministic given seeds. **pip install -r requirements.txt && python scripts/run_pipeline.py** regenerates every figure and metric on CPU in minutes; **python scripts/run_upgrade.py 1|2|3** reproduces the ensemble study; a Dockerfile provides a pinned container (`docker build -t nwm . && docker run --rm nwm`), and CI runs a smoke test of the environment, all models, and all five planners on every push.

4. Limitations and Future Work

Obstacles are dynamic but non-interactive; human-aware prediction requires interaction models (attention/GNNs) and evaluation on pedestrian datasets (ETH/UCY). The runnable core uses MLPs; included PyTorch VAE, Transformer world model, and Dreamer-style RSSM implementations provide the scaling path to raw sensors and learned stochastic latents. Perception is abstracted: obstacle states are assumed tracked. A ROS 2 node (10 Hz, `/cmd_vel`, imagined futures published for RViz) is provided for Gazebo/Isaac and hardware transfer.